

Formalization of a Proof Calculus for Incremental Linearization for NTA Satisfiability

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¹Equal contribution

Nonlinear Arithmetic and Transcendental Functions

- ▷ The NTA theory in SMT comprises nonlinear arithmetic and some transcendental functions
 - ▶ Trigonometric functions, $\exp$ and $\log$.
- ▷ Applications in formal verification of dynamic systems, digital signal processing and motion planning for robots.
 - ▶ Automation in proof assistants, if provided the necessary tooling.
- ▷ Currently, no solver with complexity better than doubly-exponential for NRA, and NTA has been shown to be undecidable.

Incremental Linearization

- ▷ *Incremental linearization* is a method introduced by Cimatti et al. for deciding the satisfiability of formulas in NTA.
 - ▶ Not a decision procedure
- ▷ Given a formula ψ in NTA, a new formula ψ' is obtained by abstracting each transcendental function and variable multiplication as an uninterpreted function.
- ▷ ψ' is thus a formula in UFLRA. If ψ' is unsatisfiable, then so is ψ .

Incremental Linearization

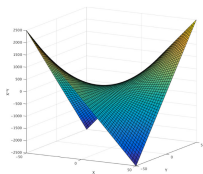
- ▷ If ψ' is satisfiable, not necessarily ψ is. In this cast, the method incrementally introduces new lemmas blocking spurious solutions.

Example

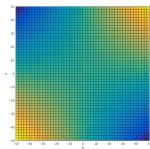
Let $\psi = x^2 + y^2 \leq 2 \wedge (x \leq -1.1 \vee x \geq 1.1) \wedge (y \leq -1.1 \vee y \geq 1.1)$. ψ' is the same, but instead of x^2 and y^2 we have $f_*(x, x)$ e $f_*(y, y)$. It is possible to interpret f_* in such a way that ψ' is satisfied, but the original formula is unsatisfiable.

Incremental Linearization

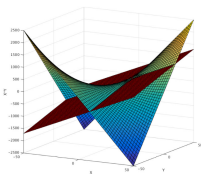
- ▶ Spurious solutions involving variable multiplication are blocked through the *tangent plane* equation and some sign properties.
- ▶ Spurious solutions involving transcendental functions are blocked using their *Taylor polynomial*.



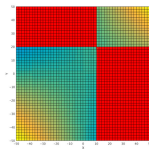
(a) $x * y$



(b) $x * y$ (top view)



(c) $x * y$ and tangent plane



(d) $x * y$ and tangent plane (top view)

Source: Cimatti et al.

Incremental Linearization

```
1: function INCREMENTALLINEARIZATION( $\psi$ )
2:    $\psi' \leftarrow \text{INITIALABSTRACTION}(\psi)$ 
3:    $\Gamma \leftarrow \emptyset$ 
4:   while INTIMELIMIT() do
5:      $\langle sat, \mu \rangle \leftarrow \text{SOLVEUFLRA}(\psi' \wedge \Gamma)$ 
6:     if not  $sat$  then return UNSAT
7:      $\langle sat', \Gamma' \rangle \leftarrow \text{CHECKREFINE}(\psi', \mu)$ 
8:     if  $sat'$  then return SAT
9:      $\Gamma \leftarrow \Gamma \cup \Gamma'$ 
```

- ▷ cvc5 implements a slightly modified version of this algorithm.
- ▷ The solver also defines a *proof calculus* composed of 21 lemmas capturing its reasoning.
- ▷ These lemmas are, for the most part, the rules that eliminate spurious solutions.

Other transcendental functions

- ▷ $\sin$ and $\exp$ are the only ones directly handled.
- ▷ $\cos(x) = \sin(x + \frac{\pi}{2})$
- ▷ $\tan(x) = \frac{\sin(x)}{\cos(x)}$
- ▷ $\sec(x) = \frac{1}{\cos(x)}$
- ▷ $\log$ and inverse trigonometric functions are modeled with new variables. For instance, if the input formula has the term $\log(1 + x)$ we introduce a new variable y and add the assertion $\exp(y) = 1 + x$.

Contributions

- ▷ In this work we mechanize proofs of all the lemmas in the proof calculus using the Lean proof assistant.
 - ▶ *lean-smt*
- ▷ In this process, we identified missing hypothesis in its documentation, as well as some other inaccuracies.
- ▷ This is the first step towards reconstructing proofs of this algorithm in Lean.

Formalization of the Proof Calculus

Lemmas on multiplication - ARITH_MULT_TANGENT

$$\frac{- \mid x, y, a, b}{xy \leq bx + ay - ab \leftrightarrow ((x \leq a \wedge y \geq b) \vee (x \geq a \wedge y \leq b))}$$

$$\frac{- \mid x, y, a, b}{xy \geq bx + ay - ab \leftrightarrow ((x \leq a \wedge y \leq b) \vee (x \geq a \wedge y \geq b))}$$

Lemmas on multiplication - ARITH_MULT_TANGENT

theorem arithMulTangentLower $\{\alpha : \text{Type}\}$ [LinearOrderedField α] (x y a b : α) :
 $x * y \leq b * x + a * y - a * b \leftrightarrow ((x \leq a \wedge y \geq b) \vee (x \geq a \wedge y \leq b))$

theorem arithMulTangentUpper $\{\alpha : \text{Type}\}$ [LinearOrderedField α] (x y a b : α) :
 $x * y \geq b * x + a * y - a * b \leftrightarrow ((x \leq a \wedge y \leq b) \vee (x \geq a \wedge y \geq b))$

$$\frac{f_1, \dots, f_k \mid m}{m \bowtie 0}$$

Each f_i has the form $x_j \bowtie 0$. m is a monomial composed by powers of the x_i 's. The relation in the conclusion is defined based on the parity of the exponents and on the sign of each x_i .

Lemmas on multiplication - ARITH_MULT_SIGN

theorem powNegOdd : $\forall \{k : \mathbb{N}\} \{r : \mathbb{R}\}, r < 0 \rightarrow \text{Odd } k \rightarrow r ^ k < 0$
theorem powNegEven : $\forall \{k : \mathbb{N}\} \{r : \mathbb{R}\}, r < 0 \rightarrow \text{Even } k \rightarrow r ^ k > 0$
theorem nonZeroEvenPow : $\forall \{k : \mathbb{N}\} \{r : \mathbb{R}\}, r \neq 0 \rightarrow \text{Even } k \rightarrow r ^ k > 0$
theorem powPos : $\forall \{k : \mathbb{N}\} \{r : \mathbb{R}\}, r > 0 \rightarrow r ^ k > 0$

theorem combineSigns₁ : $\forall \{a \ b : \mathbb{R}\}, a > 0 \rightarrow b > 0 \rightarrow b * a > 0$
theorem combineSigns₂ : $\forall \{a \ b : \mathbb{R}\}, a > 0 \rightarrow b < 0 \rightarrow b * a < 0$
theorem combineSigns₃ : $\forall \{a \ b : \mathbb{R}\}, a < 0 \rightarrow b > 0 \rightarrow b * a < 0$
theorem combineSigns₄ : $\forall \{a \ b : \mathbb{R}\}, a < 0 \rightarrow b < 0 \rightarrow b * a > 0$

Lemmas for transcendental functions - Bounding $\exp$ e $\sin$

$$\frac{- \mid d, c, t}{t \geq c \rightarrow \exp(t) \geq \text{maclaurin}(\exp, d, c)}$$

Where d is an odd number.

$$\frac{- \mid d, t, l, u}{t \in [l, u] \rightarrow \sin(t) \geq \text{secant}(\sin, l, u, t)}$$

Where d is congruent to 3 modulo 4*, $\sin$ is concave in the interval $[l, u]^*$ and secant is the secant line to the d -th Taylor polynomial at the points $(l, p(l))$ and $(r, p(r))$.

Taylor's theorem and Lagrange's remainder

- ▷ These results follow from *Taylor's theorem* with the Lagrange form of the remainder:

$$f(x) - \left(\sum_{j=0}^n \frac{f^{(j)}(x_0)}{j!} (x - x_0)^j \right) = \frac{f^{(n+1)}(x')}{(n+1)!} (x - x_0)^{n+1}$$

- ▷ This formula already existed in mathlib, but with the additional hypothesis that $x_0 < x$. We extended it to the general version.

Lemmas for transcendental functions - Bounding $\exp$ e $\sin$

```
theorem arithTransExpApproxBelow (d n :  $\mathbb{N}$ ) ( $\_ : d = 2*n + 1$ ) :  
  Real.exp x  $\geq$  taylorWithinEval Real.exp d Set.univ 0 x
```

```
theorem arithTransSineApproxBelowPos (d k :  $\mathbb{N}$ ) (hd :  $d = 4 * k + 3$ ) (t l u :  $\mathbb{R}$ )  
  (ht :  $l \leq t \wedge t \leq u$ )  
  (hl :  $0 < l$ ) (hu :  $u \leq \text{Real.pi}$ ) :  
  let p :  $\mathbb{R} \rightarrow \mathbb{R} := \text{taylorWithinEval Real.sin d Set.univ 0}$   
  sin t  $\geq ((p\ l - p\ u) / (l - u)) * (t - l) + p\ l$ 
```

Future Work

- ▷ Proof reconstruction using *lean-smt*
- ▷ Challenges with real numbers